

POWERBOX · DESIGN PRECIS

PowerBox design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	PBX-PRC-001	Version	0.2
Status	Draft	Date	2026-09-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

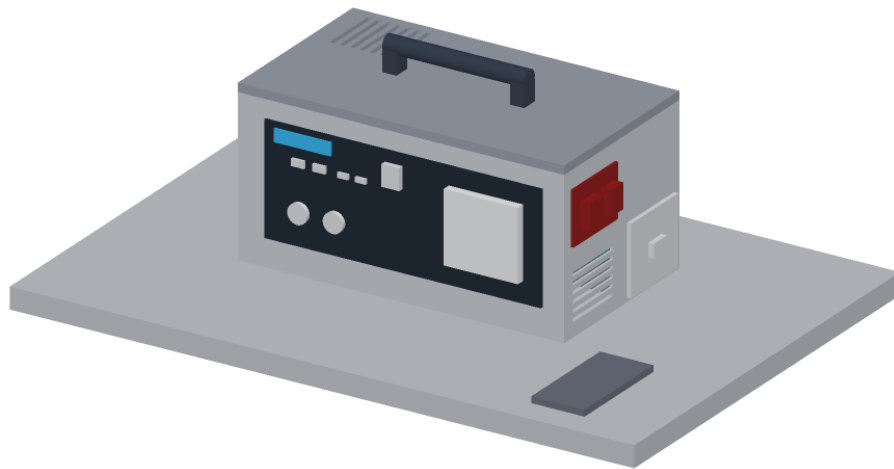
Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, input and output specification, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

PowerBox design precis

PowerBox is an aluminium carry case about the size of a small toolbox (460 x 260 x 220 mm body) with a SwapCell pack sliding into a bay through a side door. A charge controller takes DC from StepGen or a solar panel, a certified external charger takes grid power, and a charged pack from a SunSpoke bike can simply be swapped in. Power leaves through USB-C PD, USB-A, 12 V sockets and one 300 W pure sine AC outlet protected by a GFCI or RCD. It never connects to household wiring. First-order numbers suggest one pack gives about 420 Wh usable, enough for about 1.9 evenings of lights, phones, radio and a router, and the PowerBox parts cost about \$435 without the pack.

PowerBox

CONCEPT, NOT FOR FABRICATION



Grey: Table top, phone for scale

Figure 1. PowerBox on a table, with a phone for scale. Output panel on the front, input panel and pack bay door on the right end. Massing model.

■ How it works

1. **Charge.** Energy comes in through one of three paths, all controlled by the host controller: - **DC input** (StepGen or a solar panel, 12 to 60 V, up to 200 W) goes through a buck-boost charge controller that tracks the source's maximum power point and charges the pack at up to 54.6 V. - **Grid** goes through a certified external 54.6 V, 5 A charger brick with a DC plug. PowerBox itself has no AC inlet, so it cannot be plugged into anything that would back-feed. - **SunSpoke** charges the pack while riding or at its own 100 W panel. The rider brings the charged

SwapCell pack home and swaps it in, which takes about 30 s. SunSpoke's panel can also plug into the DC input directly.

2. **Store.** The SwapCell pack (13S2P lithium-ion, 46.8 V nominal, about 468 Wh) sits in a bay with guide rails and mates through the SwapCell blind-mate connector. The host controller runs the SwapCell CAN handshake: the pack enables its output only when seated, with the interlock closed and a valid host heartbeat.
3. **Convert and deliver.** The 39 to 54.6 V bus feeds a 48 V to 12 V buck converter for the 12 V sockets and the USB-C PD and USB-A modules, and a 48 V input, 300 W pure sine inverter for the single AC outlet.
4. **Inform.** A small display shows state of charge, input and output power and estimated time remaining, read from the pack's CAN messages. The host switches the inverter off after 10 min below 5 W, because its idle draw would otherwise drain the pack.
5. **Protect.** A pack fuse, a breaker on the inverter feed, per-output fuses, thermostatic fan control and the pack's own BMS limits guard against faults.

PowerBox: energy per usable cycle, DC input to loads (all values are estimates)

CONCEPT, NOT FOR FABRICATION

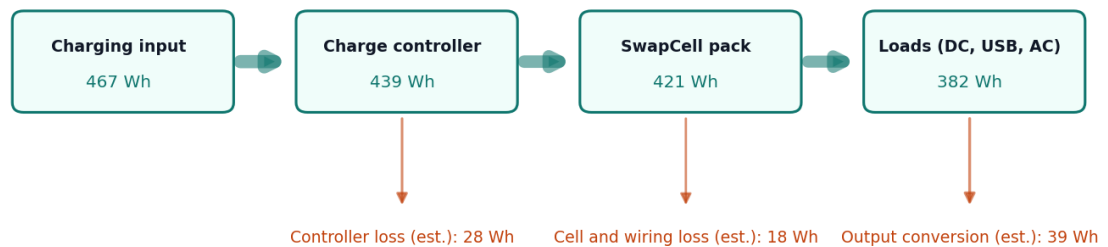


Figure 2. Energy per usable cycle, DC input to loads. All values are estimates: controller 94 %, cell charge and wiring 96 %, output conversion about 91 % for a mixed load of 60 % DC, 25 % USB and 15 % AC.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Proposed choice	Notes
1	Enclosure body	Folded 1.2 mm aluminium tub, 460 x 260 x 220 mm	Spreads heat and resists fire better than plastic. Proposed, awaiting Amish
2	Lid	Aluminium with exhaust slots at the back	Removable for service
3	Carry handle	Folding bar handle on the lid	Rated 15 kg or more
4	SwapCell pack	SwapCell interface v0.2 reference pack, 468 Wh	Not in the PowerBox cost; shared with SunSpoke and the dock
5	Pack bay	Floor, guide rails, SwapCell dock-side receptacle on a floating mount, shelf above	Mirrors the SwapCell dock cradle, horizontal
6	Pack bay door	Hinged door on the right end, magnetic catch, padlock eye	Keeps fingers and debris out of the bay
7	Multi-input charge controller	Buck-boost DC-DC, 12 to 60 V in, 200 W, CC-CV output set by the host	Maximum power point tracking in host firmware
8	Host controller	ESP32 with CAN transceiver, relays and current sensors	Runs the SwapCell handshake, charge control, auto-off and display
9	Inverter	300 W pure sine, 48 V input, 600 W surge, certified	Regional variant (230 V or 120 V) awaiting Amish
10	DC-DC converter	48 V to 12 V, 20 A buck	Feeds 12 V sockets and USB modules
11	Output panel	USB-C PD 100 W and 60 W, 2 x USB-A, 2 x 12 V sockets, main switch	Front face
12	AC outlet	Single outlet behind a 30 mA RCD (230 V) or 5 mA GFCI (120 V)	Only mains-voltage point on the box
13	Display	2.4 in TFT or e-paper	State of charge, power in and out, time left
14	Input panel	2 x Anderson Powerpole PP45: DC in, charger in	Right end, above the bay door
15	Exhaust fan	80 mm 12 V fan, thermostatic; filtered intake slots on the right end	Pulls air front to back over the inverter

Items 16 (protection and wiring), 17 (grid charger brick) and 18 (hardware) are in the BOM but not modelled.

PowerBox: cutaway

CONCEPT, NOT FOR FABRICATION

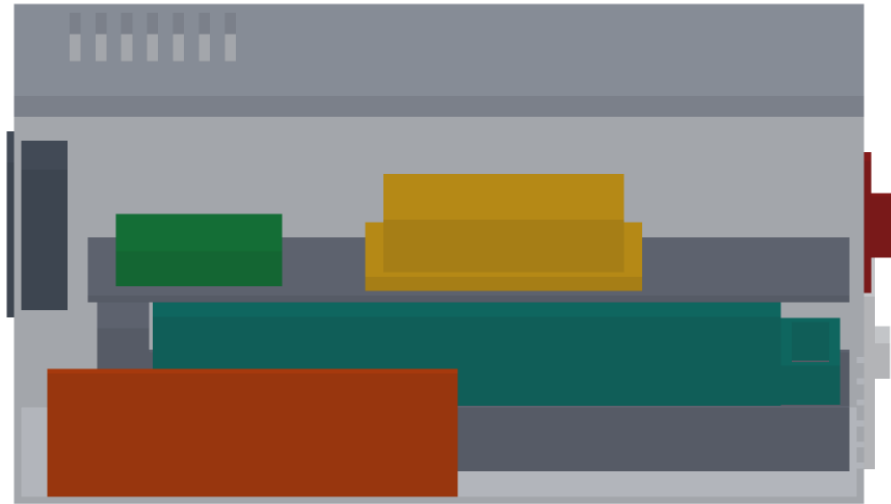


Figure 3. Section looking from the front: inverter (9) on the floor at the front, SwapCell pack (4) in its bay at the back, charge controller (7) and host (8) on the shelf above, fan (15) on the left end, input panel (14) and bay door (6) on the right end.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3. Assumptions: the SwapCell reference pack (468 Wh, 39.0 to 54.6 V); the host allows 10 to 100 % state of charge; 12 V buck 94 %, USB modules 93 % from 12 V, charge controller 94 %, cell charge and wiring 96 %, grid charger 90 %; inverter efficiency and idle draw typical of 48 V, 300 W pure sine units (not yet from a datasheet).

Table 2. Energy, charge times, size and cost.

Quantity	Estimate	Basis	Requirement
Usable energy	about 421 Wh	90 % of 468 Wh (10 to 100 %)	R1 met (400 Wh)
Reference evening (Table 1 of PBX-REQ-001)	225 Wh from the pack	155 Wh DC at 94 %, 48 Wh USB at 87 %, 5 Wh host and display over 5 h	
Evenings per pack	about 1.9	421 / 225	R2 not met (target 2)
Grid charge, 10 to 100 %	about 2.1 h; about 490 Wh from the wall	439 Wh into the pack at 5 A (about 270 W) in CC, then CV; charger 90 %	R4 met (3 h)
Solar, 200 W panel	about 635 Wh per clear day; full in one day, or about 3 h of strong sun	4.5 peak sun hours, 0.75 derating for heat, dust and angle, 94 % controller	R5 met
Solar, SunSpoke 100 W panel	about 315 Wh per clear day; one evening's use in about 0.7 day	Same basis	
StepGen, per hour of stepping	about 55 to 90 Wh stored	StepGen's estimate of 60 to 100 Wh at its output, times 94 % and 96 %	
StepGen, one evening's use	about 2.5 to 4 h of stepping	225 Wh at 55 to 90 Wh per hour	
StepGen, 10 to 100 %	about 5 to 8 h of stepping; about 12 h at 40 W	421 Wh at 55 to 90 Wh per hour	
SunSpoke pack swap	about 30 s	Open door, pull pack by its handle, insert charged pack, close door	R12 met, to confirm with users
Inverter efficiency	about 88 % at 200 to 300 W, 85 % at 60 W, 75 % at 20 W; idle draw 6 to 10 W	Typical 48 V, 300 W units	
Inverter idle, if left on	about 190 Wh per day, nearly half the pack	8 W for 24 h	Auto-off after 10 min below 5 W (R8)
Standby, off	about 3 % per month	Li-ion self-discharge about 2 to 3 % per month; BMS sleep a few mW	R8 met (5 %)
Standby, ready (display on, outputs off)	about 0.6 W	ESP32 modem sleep 0.2 W, CAN and BMS awake 0.2 W, display 0.2 W	R8 met (1.0 W); still about 14 Wh per day, so the host returns to off after 30 min idle
Standby, DC outputs on, no load	about 1.2 W	Ready plus 12 V buck and USB module quiescent draw	
Peak pack current	about 12 A	400 W output limit at 88 % and 39 V	Within SwapCell's 20 A continuous
Heat at full AC load	about 41 W from the inverter, plus about 12 W from the	300 W at 88 %; 200 W at 94 %	Fan needed

Quantity	Estimate	Basis	Requirement
	controller when charging at 200 W		
Mass	about 8.3 kg (18 lb) with pack	Pack 2.8, body 1.4, lid 0.4, handle 0.2, inverter 1.2, bay 0.4, controller 0.3, DC-DC 0.3, panels 0.4, wiring and fuses 0.6, fan and host 0.3 kg	R10 met (10 kg); charger brick about 0.8 kg extra
Size	460 x 260 x 220 mm body; about 482 x 276 x 278 mm overall	Pack 340 mm plus 35 mm handle zone, receptacle and walls set the length	R10 met (500 x 300 x 280 mm), tight on height
PowerBox parts cost	about \$435	Indicative prices, see bom/bom.csv	R11 met, about \$15 margin
Cost with one SwapCell pack	about \$805	\$435 plus about \$370 from the SwapCell BOM	Above the \$450 budget; see design choices

Table 3. Runtime on one full pack (421 Wh usable) for single loads. Estimates.

Load	Output	Power at the load	Drawn from the pack	Runtime
Wi-Fi router and fibre terminal	12 V DC	12 W	12.8 W	about 33 h
Three 5 W LED bulbs	12 V DC	15 W	16.0 W	about 26 h
Radio	12 V DC	5 W	5.3 W	about 79 h
Phone charges	USB	12 Wh each	13.7 Wh each	about 30 charges
Laptop	USB-C PD	45 W	51 W	about 8 h
Small LED TV	AC	60 W	71 W	about 6 h
Desk fan	AC	25 W	32 W	about 13 h
Maximum AC load	AC	300 W	341 W	about 1.2 h
Refrigerator or kettle	AC	Out of scope: starting surge or power beyond the 300 W inverter		

Table 4. Input and output specification (proposed).

Port	Type and connector	Voltage	Power or current	Notes
DC in	Anderson PP45, red and black	12 to 60 V DC	200 W maximum, about 15 A at 12 V	StepGen, solar panel or SunSpoke panel; MPPT; TVS and reverse-polarity protection; one source at a time
Charger in	Anderson PP45, keyed differently from DC in	54.6 V DC from the certified brick	5 A	Grid charging only through the external charger
Pack	SwapCell blind-mate connector	39.0 to 54.6 V	20 A continuous available; PowerBox uses 12 A or less	CAN 2.0B at 250 kbit/s; 120 Ω termination in PowerBox
USB-C PD 1	USB-C	5 to 20 V	100 W	Laptops
USB-C PD 2	USB-C	5 to 20 V	60 W	Phones, tablets
USB-A 1, 2	USB-A	5 V	12 W each	Phones, lamps, radios
12 V DC 1	Car socket	12 V regulated	10 A shared with 12 V DC 2	Router, lights, radio
12 V DC 2	5.5 x 2.1 mm barrel	12 V regulated	As above	
AC out	One regional socket behind a GFCI or RCD	230 V 50 Hz or 120 V 60 Hz, pure sine	300 W continuous, 600 W for 1 s	Auto-off; never to be connected to household wiring
Total output			400 W	Host limit to keep pack current and heat in range

■ Key design choices

All are proposed, awaiting Amish.

- **Removable SwapCell pack, with a fixed internal pack as fallback.** A removable SwapCell pack is the point of the design: one battery moves between PowerBox, SunSpoke and the SwapCell dock, and a worn pack is replaced without discarding the box. The fallback is a fixed internal 12.8 V LiFePO₄ pack (about 30 Ah, 384 Wh, about \$110), with a 12 V inverter and no CAN. It is safer chemistry, cheaper, fits the \$450 budget with the pack included and does not depend on SwapCell's schedule, but it loses swapping and sharing. Recommendation: removable SwapCell pack, with the fixed-pack variant kept as a documented fallback.
- **Side-loading horizontal bay.** A vertical, top-loading bay would make the case about 450 mm tall. Laying the pack along the case length with a door on the end keeps the case low and stable. Recommendation: horizontal side-loading bay.
- **One DC input with maximum power point tracking.** A single buck-boost input covers StepGen, solar and SunSpoke's panel. Two independent inputs would allow solar and StepGen at the same time, for about \$30 more. Recommendation: one DC input for the first build.

- **Grid charging only through a certified external charger.** Keeps mains-input electronics out of the box, reuses the SwapCell dock charger class, and means the box has no AC inlet. Recommendation: external charger brick.
- **A 300 W AC outlet with a DC-only variant.** AC is what users expect, but it brings mains voltage into the home, costs about \$92 and wastes energy at idle. Recommendation: include 300 W AC with auto-off, and document a DC-only variant for lights, phones and routers.
- **Aluminium enclosure.** Aluminium spreads heat and delays a fire better than a rugged plastic case, at similar cost if folded locally. Recommendation: folded aluminium.
- **Pass-through operation.** Users will want to charge from solar while running the router. The SwapCell v0.2 behavior rules allow charge only with a dock heartbeat and do not define a host that charges and discharges at once. Recommendation: ask SwapCell for a "station" host type with a combined charge and discharge mode, rather than work around it locally.
- **Budget.** Recommendation: keep \$450 for the PowerBox parts excluding the pack, and report the with-pack cost (about \$805) openly. Alternative: raise the budget to about \$850 to include one pack. Proposed, awaiting Amish.

PowerBox: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Enclosure body
- 2 Lid with exhaust slots
- 3 Carry handle
- 4 SwapCell pack (reference)
- 5 Pack bay and blind-mate receptacle
- 6 Pack bay door
- 7 Multi-input charge controller
- 8 Host controller (ESP32, CAN)
- 9 300 W pure sine inverter
- 10 DC-DC converter, 48 V to 12 V
- 11 Output panel (USB-C PD, USB-A, 12 V)
- 12 AC outlet with GFCI
- 13 State-of-charge display
- 14 Input panel (DC in, grid charger in)
- 15 Exhaust fan and grille

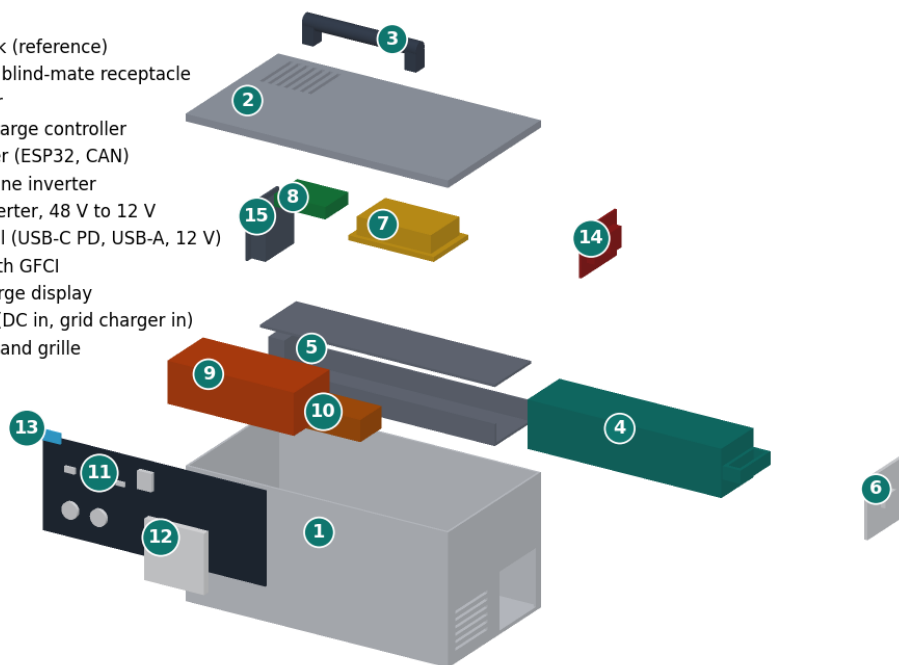


Figure 4. Exploded view with BOM numbers.

■ Safety

Safety: PowerBox is standalone only. Never connect the AC outlet to a wall outlet, household wiring, a distribution board or a transfer switch. Back-feeding can kill utility workers repairing lines during an outage, can damage the box and the home's wiring, and is illegal.

Safety: The SwapCell pack is a 468 Wh lithium-ion battery. Use it only with the SwapCell BMS, keep the 40 A pack fuse in place, and never charge a damaged, swollen or wet pack. Charge on a non-combustible surface, away from exits, beds and children, and do not cover the box while charging. Keep a smoke alarm in the room.

Safety: The AC outlet carries mains voltage (230 V or 120 V). It must be behind a GFCI or RCD, and for the protection to work the inverter's output neutral must be bonded to the case and the protective earth pin, as in a vehicle or boat installation. Many small inverters have a floating output where a GFCI will not trip. This must be confirmed for the chosen inverter at TRL 3.

- **No back-feed by design.** There is no AC inlet on the box, and grid charging uses a certified external charger with a DC plug, so no cable can join PowerBox's AC output to a live circuit through an ordinary plug. The user guide and labels must still warn against double-male cords and improvised connections.
- **Extra-low-voltage bus.** The battery side stays at 54.6 V or less, below the 60 V DC limit for extra-low voltage, so only the inverter output and the charger brick input are at mains voltage.
- **Ventilation.** The inverter can shed about 41 W and the controller about 12 W. A thermostatic fan pulls air through filtered intake slots on the right end and out through the left end and lid. The host derates or shuts outputs down if internal temperature exceeds a limit, and the pack enforces its own temperature limits. The case must not be used in a closed cupboard or bag.
- **Cords and connectors.** Keyed Powerpole connectors prevent the charger and DC inputs from being swapped. Input and output cords must be rated for their current, kept out of walkways and never run under rugs. The 12 V car socket is fused at 10 A.
- **Pack handling.** The bay door keeps fingers away from the connector, and the SwapCell interlock keeps the pack output dead until it is fully seated.
- **Human power.** StepGen is moving machinery with its own safety section in its design; PowerBox limits input current so a generator cannot over-charge the pack.
- **Not a medical or life-safety supply.** PowerBox must not be relied on for oxygen concentrators or other life-support equipment.

■ Open questions for TRL 3

- **Waking the pack from a battery-only host.** SwapCell's WAKE pin needs 5 to 15 V from the host, but with the pack asleep and no charger connected, PowerBox has no other source. Options: a small keep-alive cell in PowerBox, SwapCell waking on interlock closure, or a wake button on the pack. Raise with SwapCell; do not change the interface locally.
- **Pass-through charging** needs a SwapCell host mode that allows charge and discharge together (see design choices).
- Confirm inverter idle draw, efficiency at light load and whether its output neutral can be bonded for GFCI or RCD operation.

- Confirm that the buck-boost controller stays at or above 90 % efficiency at 40 W and tracks StepGen's varying voltage without hunting (R6, at risk).
- Choose the AC region first (230 V 50 Hz or 120 V 60 Hz). Proposed: 230 V, awaiting Amish.
- The evening profile misses R2 by about 5 %. Options: accept 1.9 evenings, lower the discharge cut-off to 5 % (about 445 Wh, 2.0 evenings but more cell wear), or revise the reference profile with users.
- Where to place the charge controller relative to the pack, so its heat does not warm the cells.
- Should the display count energy delivered, for a community charging point that charges per phone?
- Validate the load profile, pack-swap routine and price with users through a local partner.

Concept media: [blueprint sheet](#), [interactive 3D model](#).